Test Plan Report

1. Introduction

This test plan describes the testing process for the Manifetch NICU Monitoring System. The system is a web-based clinical decision support platform that monitors neonatal physiological signals in real time and generates AI-assisted risk alerts. Our goal is to verify that the system meets its requirements and is consistent with the architectural decisions made in the HLD and LLD reports.

The test scope covers all packages defined in the LLD: the Data Simulation layer (Simulator, ScenarioConfig, StreamPublisher), the Backend (DataIngestionController, AlertService, RuleEngineService and related model classes), the AI Module (InferenceController, InferenceService, ModelRunner), and the Frontend (DashboardPage, WebSocketClient, ApiClient). As part of the AI inference testing, Random Forest and XGBoost models will be evaluated comparatively with SHAP-based explainability.

All tests are conducted using PICS-DB (PhysioNet) and synthetic data generated via NeuroKit2. In compliance with KVKK and GDPR regulations, no real patient data is used at any stage.

2. Project Scope

Project Title:

A Unified Web-Based Platform for Real-Time NICU Signal Monitoring

Brief Description of the System:

Manifetch is a secure web-based platform that processes simulated neonatal physiological signals (ECG, SpO_2, heart rate, respiration rate) and delivers real-time risk scores and clinical alerts to authorized healthcare staff within a single-hospital environment.
Each AI prediction includes a SHAP explanation so that clinicians can understand why the model made a particular decision. The system runs entirely on-premises and does not replace certified medical devices.

Main Functionalities of the System:

- Real-time visualization of neonatal physiological signals (ECG, SpO2, heart rate, respiration rate) via WebSocket streaming
- Synthetic neonatal signal generation using NeuroKit2 across four clinical scenarios (normal, sepsis, apnea, cardiac anomaly)
- Z-score normalization of physiological signals

- Multi-label AI risk scoring using Random Forest and XGBoost models for sepsis, apnea, and cardiac anomaly detection
- SHAP-based explainability showing top contributing features for each AI prediction
- Rule-based threshold evaluation (RuleEngineService) with patient-specific ThresholdRule configuration
- Alert lifecycle management: ACTIVE → ACKNOWLEDGED → RESOLVED (AlertService)
- Role-based access control (RBAC) for doctors, nurses, and administrators (User model)
- Historical trend visualization and patient record management
- PDF clinical report export for discharged or monitored patients

3. Features to Be Tested

The following features are in scope for testing, mapped to their corresponding LLD components:

Feature 1: Signal normalization and threshold calibration

Feature 2: Synthetic neonatal data generation across clinical scenarios

Feature 3: Feature extraction from ECG, SpO2, heart rate, and respiration rate signals

Feature 4: Random Forest and XGBoost multi-label risk classification with model comparison

Feature 5: SHAP value computation and top-feature identification per prediction

Feature 6: Rule-based alert triggering and patient-specific threshold evaluation

Feature 7: Alert lifecycle management and acknowledgement

Feature 8: Real-time WebSocket data streaming from simulation to dashboard

Feature 9: Role-based access control for all three user roles

Feature 10: PDF clinical report generation

Feature 11: Historical trend visualization

Feature 12: Patient record creation and management

Feature 13: Concurrent stream stability with 10 simultaneous patient streams

4. Testing Methodology

The following testing methods are used in this project:

Unit Testing: The classes and functions defined in the LLD are tested in isolation. The goal of these tests is to verify that each component works correctly on its own. The test scope includes the signal normalization pipeline (DataIngestionController), feature extraction functions in InferenceService and ModelRunner, rule-based label assignment in RuleEngineService, SHAP value computation, alert level determination in AlertService, and ThresholdRule evaluation.

Integration Testing: The data flow between modules is tested according to the package interaction structure defined in the LLD. The main paths tested are: signal streaming from StreamPublisher to DataIngestionController via WebSocket, request forwarding from DataIngestionController to InferenceController via REST, delivery of AIResult to AlertService, and finally data flow to DashboardPage through DashboardController and WebSocketClient. The JSON format is verified at each module boundary.

System Testing: Four clinical scenarios are tested end-to-end using synthetic data: a normal scenario where no alert is expected, a sepsis scenario where heart rate rises and SpO2 drops, an apnea scenario where breathing stops for more than 20 seconds, and a cardiac anomaly scenario where ECG irregularities are observed. Alert lifecycle transitions and RBAC restrictions are also validated at this stage.

Performance Testing: The system is evaluated against the non-functional requirements defined in the Project Specifications Report. Real-time signal updates must not exceed 1 second (NFR-2), AI inference for 10 concurrent patients must not exceed 3 seconds (NFR-2), and the system must remain stable under 10 simultaneous WebSocket streams (NFR-4). Random Forest and XGBoost are compared on a patient-level holdout test set; the target F1 score is 0.80 or above for each condition. SHAP computation time is also measured within the inference budget.

User Testing: Realistic clinical workflows are simulated to evaluate usability (NFR-3). In the nurse scenario, it is tested whether a clinician can locate and acknowledge a critical alert within 10 seconds of it appearing on DashboardPage. In the doctor scenario, it is checked whether the SHAP explanation panel correctly identifies the most contributing clinical feature. In the historical trend scenario, it is tested whether the doctor can correctly interpret a 7-day signal trend.

5. Test Environment

Hardware:

- PC with at least 8 GB RAM

- Optional: GPU for model training

Software:

- Operating System: Windows 11 / Ubuntu 24
- Backend: FastAPI (Python 3.10+)
- Frontend: React.js
- Real-Time Communication: WebSocket (RFC 6455)
- AI / ML: scikit-learn (Random Forest), XGBoost, SHAP (TreeExplainer)
- Signal Processing and Synthetic Data: NeuroKit2, SciPy, PyWavelets
- Dataset: PICS-DB (PhysioNet) — ECG, respiration rate, heart rate; SpO_2 is generated synthetically via NeuroKit2
- Database: SQLite
- Security: TLS 1.3 is used for data transmission, AES-256 for data storage

6. Test Cases

Each test case has a unique Test ID. The Result column will be filled in after the tests are executed.

6.1 Unit Tests

Test ID	Description	Input(s)	Expected Output	Result
TC-01	Signal normalization	Raw ECG array, gestational age 28 weeks	Z-score normalized ECG array	
TC-02	Threshold calibration - term newborn	Heart rate 160, chronological age 40 days	Threshold correctly mapped to 100–160 range	
TC-03	Threshold calibration – preterm	Heart rate 165, gestational age 26 weeks	Threshold correctly mapped to 120–170 range	
TC-04	ECG feature extraction	30-second normalized ECG window	HRV, R-R interval, abnormal beat count computed	
TC-05	SpO2 feature extraction	30-second SpO2 array	Mean, minimum, time below 90% computed	
TC-06	Respiration rate feature extraction	30-second respiration rate array	Mean, apnea duration, variability computed	
TC-07	Heart rate feature extraction	30-second heart rate array	Mean, max/min, sudden change computed	

TC-08	Synthetic data generation - apnea	ScenarioConfig: apnea, gestational age 32 weeks	Respiration stopped for more than 20 seconds	
TC-09	Synthetic data generation - sepsis	ScenarioConfig: sepsis, term newborn	Heart rate >180, SpO2 <90%	
TC-10	Rule-based label assignment	Heart rate 185, SpO2 88%, respiration 65	Sepsis suspect label assigned	
TC-11	SHAP value computation	15-feature vector, trained model	Top 3 contributing features identified	
TC-12	Alert level determination	Sepsis risk score 0.78	Alert level set to HIGH	
TC-13	Threshold rule evaluation	SpO2 88%, minimum threshold 90%	Rule violation detected, alert triggered	
TC-14	RF and XGBoost comparison	Same feature vector fed to both models	Both models produced risk scores, results saved	

6.2 Integration Tests

Test ID	Description	Input(s)	Expected Output	Result
TC-15	WebSocket streaming from simulation to backend	Synthetic signal stream from Simulator	DataIngestionController received and processed the data	
TC-16	Request from data ingestion to AI module	Normalized signal JSON	InferenceController received the request, AIResult produced	
TC-17	AI module to notification service	Sepsis risk score 0.78, alert level HIGH	AlertService created an alert object	
TC-18	Notification service to dashboard	Alert object	Alert displayed correctly on dashboard	
TC-19	WebSocket real-time streaming	Continuous signal stream	Dashboard updated within 1 second	
TC-20	AIResult linked to patient record	AIResult with risk scores and SHAP values	AIResult associated with patient record in database	

6.3 System Tests

Test ID	Description	Input(s)	Expected Output	Result

TC-21	Normal scenario end-to-end	Heart rate 140, SpO2 97%, respiration 45 — term newborn	No alert generated, signals displayed on dashboard	
TC-22	Sepsis scenario end-to-end	Heart rate rising to 190, SpO2 dropping to 87%	HIGH sepsis alert generated, SHAP explanation displayed	
TC-23	Apnea scenario end-to-end	Respiration stops for more than 20 seconds — preterm patient	HIGH apnea alert generated within 1 second	
TC-24	Cardiac anomaly scenario end-to-end	Sudden heart rate fluctuation in ECG — 1–3 month patient	Cardiac anomaly alert generated, SHAP explanation displayed	
TC-25	Alert acknowledgement	Clinician acknowledges HIGH alert	Alert status changed from ACTIVE to ACKNOWLEDGED	
TC-26	Alert resolution	Clinician resolves acknowledged alert	Alert status changed from ACKNOWLEDGED to RESOLVED	
TC-27	RBAC — doctor login	Valid doctor credentials	Doctor dashboard accessible, admin panel restricted	
TC-28	RBAC — nurse login	Valid nurse credentials	Real-time monitoring accessible, patient management restricted	
TC-29	RBAC — administrator login	Valid administrator credentials	System configuration and user management accessible	
TC-30	PDF report generation	Patient P001, 7-day date range	PDF report generated with historical signals and AI results	
TC-31	Historical trend visualization	Patient P001, last 24 hours	Historical signals displayed in trend chart	
TC-32	Patient record creation	Gestational age, chronological age, gender, birth weight	Patient record successfully created	

6.4 Performance Tests

Test ID	Description	Input(s)	Expected Output	Result

TC-33	AI inference latency	10 concurrent feature vectors, RF and XGBoost	All results returned within 3 seconds	
TC-34	Real-time update latency	10 concurrent WebSocket streams	Dashboard updates delivered within 1 second	
TC-35	Concurrent stream stability	10 simultaneous streams for 5 minutes	No data loss, system remained stable	
TC-36	RF and XGBoost accuracy comparison	Labeled test set	Both models evaluated, F1 ≥ 0.80, best model selected	
TC-37	SHAP computation time	Labeled test set	SHAP values computed within 3-second inference budget	

6.5 User Tests

Test ID	Description	Input(s)	Expected Output	Result
TC-38	Nurse — alert detection and acknowledgement	HIGH sepsis alert active on dashboard	Nurse located and acknowledged the alert within 10 seconds	
TC-39	Doctor — AI explanation interpretation	SHAP explanation panel for HIGH sepsis alert	Doctor correctly identified the most contributing feature	
TC-40	Doctor — historical trend analysis	7-day trend data for patient P001	Doctor correctly interpreted the signal trend	

7. Risks and Limitations

Possible Technical Risks:

- Synthetic signals may not fully reflect the noise and artifact characteristics of real NICU hardware, which could affect model performance on real data
- Running SHAP computation and model inference together for 10 concurrent patients may make it difficult to meet the 3-second latency constraint
- Long-running WebSocket sessions may cause memory or connection management issues
- Running both models simultaneously may lead to performance problems on low-resource hardware

Dataset and Hardware Limitations:

- Since PICS-DB does not include SpO_2, it is generated synthetically via NeuroKit2 and may differ from real SpO_2 distributions
- All tests are conducted with simulated or anonymized data in compliance with KVKK and GDPR, no real patient data is used
- Model training without a GPU may be slow, which could extend the development process

8. Conclusion

This test plan has been prepared for the systematic testing of the Manifetch NICU Monitoring System. A total of 40 test cases have been defined, covering everything from unit tests to user testing, each directly linked to the project requirements and design decisions in the LLD.

The plan directly addresses the key features of the system. Random Forest and XGBoost models will be evaluated comparatively on the same dataset, and the model achieving an F1 score of 0.80 or above will be selected for integration. SHAP explainability is tested both to verify the correctness of model outputs and to assess whether clinicians can interpret these outputs correctly.

Performance tests will verify whether the system remains stable under 10 concurrent patient streams and meets the real-time latency constraints. User tests will confirm that clinical workflows operate as expected. Upon completion of all these tests, the system will be verified as ready for the demo.

9. Requirement Traceability Matrix

Requirement **ID**	**Description**	Related Test Cases
F-1	Real-time signal display	TC-15, TC-19, TC-21, TC-34
F-2	AI-assisted risk scoring and SHAP visualization	TC-11, TC-14, TC-16, TC-22, TC-23, TC-24, TC-36, TC-39
F-3	Role-based access control	TC-27, TC-28, TC-29
F-4	Patient record management	TC-32
F-5	Alert and notification system	TC-12, TC-13, TC-17, TC-18, TC-22, TC-23, TC-24, TC-25, TC-26
F-6	PDF report export	TC-30
F-7	Historical trend visualization	TC-31, TC-40
NFR-1	Security: TLS 1.3, AES-256	TC-27, TC-28, TC-29
NFR-2	Performance: latency ≤1s, inference ≤3s	TC-33, TC-34, TC-35, TC-37

NFR-3	Usability	TC-38, TC-39, TC-40
NFR-4	Reliability: 10 concurrent streams	TC-34, TC-35
NFR-5	Maintainability / Portability	TC-35